OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T2: Systemidentifikation

Versuch 1

Ausarbeitung als Beispiel

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`Dokument-Version: 02`

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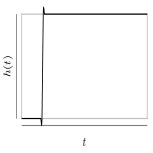
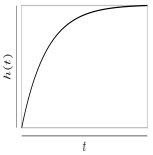
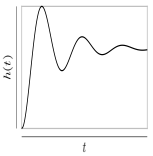
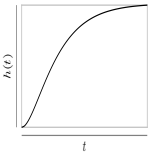
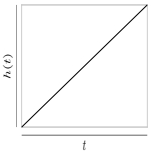
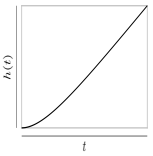
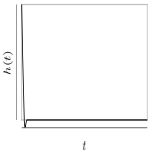
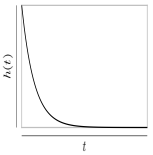
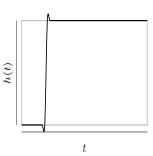
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1 Preperations

Zur Vorbereitung auf diesen Versuch wurden die in ELO bereitgestellten Dateien heruntergeladen und gespeichert [**RTP˙Versuche**]. Zudem wurden die jeweiligen Sprungantworten verschiedener Systemverhalten recherchiert:

1.1. Different basic system behaviors

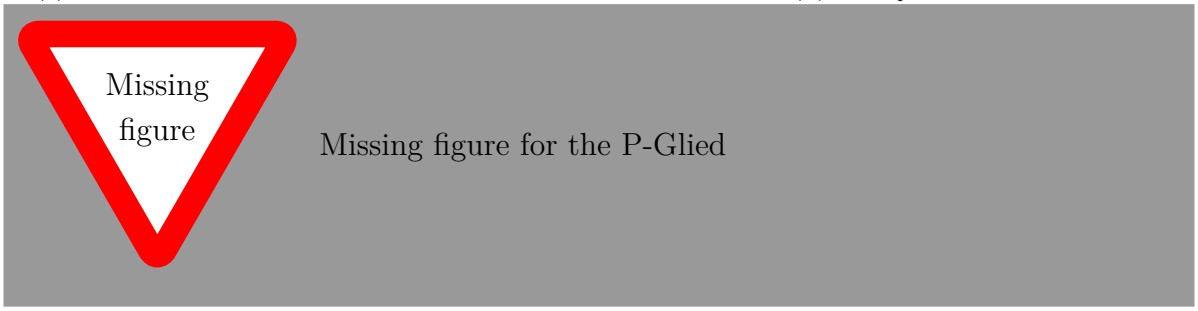
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Systemverhalten	**Übertragungsfunktion**	**Visualization**
P-Glied (K_p)	$G(s) = K_p$ (1.1)	
PT_1-Glied (K_p, T_1)	$G(s) = \frac{K_p}{T_1 s + 1}$ (1.2)	
PT_2-Glied (K_p, ω_0, D)	$G(s) = \frac{K_p}{\frac{1}{\omega_0^2} s^2 + \frac{2D}{\omega_0} s + 1}$ (1.3)	
PT_2-Glied für $D \geq 1$ (K_p, T_1, T_2)	$G(s) = \frac{K_p}{(T_1 s + 1)(T_2 s + 1)}$ (1.4)	
I-Glied (K_I)	$G(s) = \frac{K_I}{s}$ (1.5)	
IT_1-Glied (K_I, T_1)	$G(s) = \frac{K_I}{s(T_1 s + 1)}$ (1.6)	
D-Glied (K_D)	$G(s) = K_D\, s$ (1.7)	
DT_1-Glied (K_D, T_1)	$G(s) = \frac{K_D\, s}{T_1 s + 1}$ (1.8)	
T_t-Glied (K_p, T_t)	$G(s) = K_p\, e^{-T_t s}$ (1.9)	

1.1.1. P-Glied

Explanation

Proportional-Glied, show a constant behavior in steady state for a constant input. The output is directly proportional to the input, with the proportionality constant being K_p. The equation for a P-Glied is given by $x_a(t) = K_p x_e(t)$, where $x_a(t)$ is the output and $x_e(t)$ is the input. The transfer function of a P-Glied is $G(s) = K_p$.



Missing figure for the P-Glied

Parameters needed for characterization

The only parameter needed to characterize a P-Glied is the proportional gain K_p. This parameter determines the steady-state gain of the system and how much the output will change in response to a change in the input.

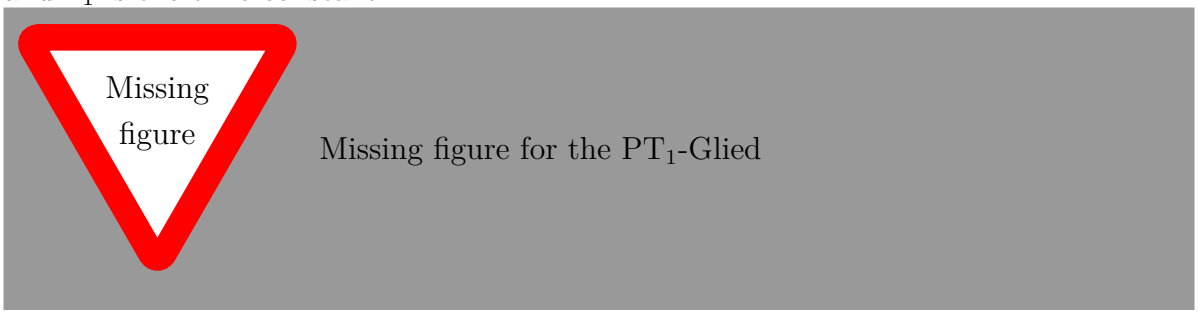
Step and Impulse response

The step response of a P-Glied is a step function that reaches a steady state value of K_p times the input step magnitude. The impulse response is an impulse function scaled by K_p.

1.1.2. PT$_1$-Glied

Explanation

The PT$_1$-Glied is a first-order delay system that consists of a proportional element and a first-order part. The transfer function of a PT$_1$-Glied is given by $T_1\dot{x}_a + x_a(t) = K_p x_e(t)$, or $G(s) = \dfrac{K_p}{T_1 s + 1}$, where K_p is the proportional gain and the final value of the function and T_1 is the time constant.



Missing figure for the PT$_1$-Glied

Parameters needed for characterization

For the characterization of a PT$_1$-Glied, the parameters needed are the proportional gain K_p and the time constant T_1. The proportional gain K_p determines the steady-state gain of the system, while the time constant T_1 determines how quickly the system responds to changes in the input. The System is also characterized by its initial tangent, which is given by $y(t) = \frac{K_p}{T_1}t$. The point at which the initial tangent intersects the steady state value of the step response can be used to determine the time constant T_1. T_1 is the time it takes for the system to reach approximately 63% of its final value in response to a step input. The threefold multiple of the time constant T_1 at which the system reaches 95% of its final value can also be used to characterize the system's response speed.

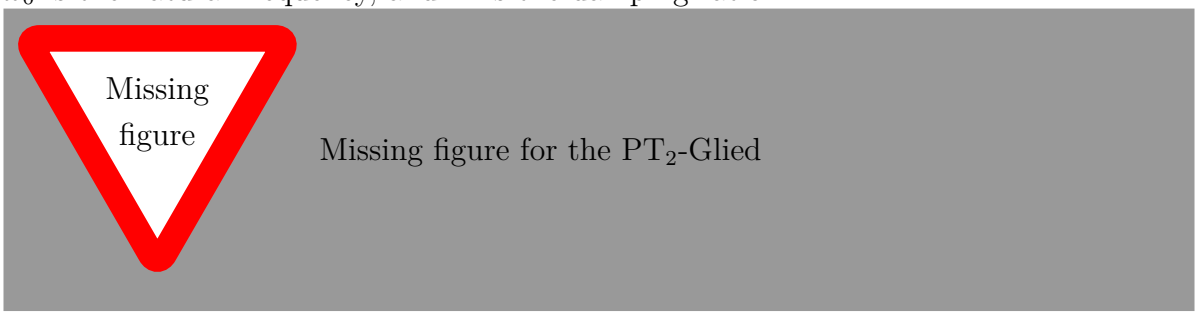
Step and Impulse response

The step response of a PT$_1$-Glied is an exponential function that approaches a steady state value of K_p times the input step magnitude, with a time constant of T_1. The impulse response is an exponential function scaled by K_p/T_1.

1.1.3. PT$_2$-Glied

Explanation

A PT$_2$-Glied is a second-order delay system that consists of a proportional element and a second-order part, which appares delayed and oscillatory behavior becuase of its second order differential nature. The transfer function of a PT$_2$-Glied is given by $\frac{1}{omega_0^2}\ddot{x}_a + \frac{2D}{\omega_0}\dot{x}_a + x_a(t) = K_p x_e(t)$, or $G(s) = \frac{K_p}{\frac{1}{\omega_0^2}s^2 + \frac{2D}{\omega_0}s + 1}$, where K_p is the proportional gain, ω_0 is the natural frequency, and D is the damping ratio.



Missing figure for the PT$_2$-Glied

Parameters needed for characterization

For the characterization of a PT$_2$-Glied, the parameters needed are the proportional gain K_p, the natural frequency ω_0, and the damping ratio D. The proportional gain K_p determines the steady-state gain of the system, while the natural frequency ω_0 and damping ratio D determine the oscillatory behavior and response speed of the system. The damping ratio D indicates whether the system is underdamped ($D < 1$), critically damped ($D = 1$), overdamped ($D > 1$) or undamped ($D = 0$).
For an underdamped system ($D < 1$), the step response will exhibit oscillations before settling to the steady state value. The natural frequency ω_0 determines the frequency of these oscillations, while the damping ratio D determines how quickly the oscillations decay. For a critically damped system ($D = 1$), the step response will reach the steady state value without oscillating, and it will do so as quickly as possible without overshooting. It is often reached by putting two PT$_1$-Glied in series, which results in the transfer function given by $T_1T_2\ddot{x}_a + (T_1 + T_2)\dot{x}_a + x_a(t) = K_p x_e(t)$, or $h(t) = K_P\left(1 - (1 + \omega_0 t)e^{-\omega_0 t}\right)$.
For an overdamped system ($D > 1$), the step response will reach the steady state value without oscillating, but it will take longer to do so than a critically damped system. The Transfer function can be rewritten as $h(t) = K_p\left(1 + \frac{1}{\lambda_1 - \lambda_2}(\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t})\right)$, where λ_1 and λ_2 are the eigenvalues of the characteristic equation of the system, given by $\lambda_{1,2} = -D\omega_0 \pm \omega_0\sqrt{D^2 - 1}$. The time constants of the system can be determined from these roots, and they indicate how quickly the system responds to changes in the input.
For an undamped system ($D = 0$), the step response will exhibit sustained oscillations at the natural frequency ω_0.
For the case $0 < D < 1$, the transfer function can be rewritten as $h(t) = K_p\left(1 - \frac{e^{-\omega_0 Dt}}{\sqrt{1-D^2}}\sin(\omega_d t + \phi)\right)$.
The Periodic time of the oscillations can be determined from the damped natural frequen-

cy $T_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_0\sqrt{1-D^2}}$. The Transferfunction can be caracterized by the upper and lower envelope of the oscillations, which are given by $h^{\pm}(t) = K_p(1 \pm \frac{e^- D\omega_0 t}{\sqrt{1-D^2}})$. The over swing time t_{max}, which is the time at which the first maximum of the step response occurs, can be determined from the equation $t_{max} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_0\sqrt{1-D^2}}$. The overshoot h_{max}, can be determined from the equation $h_{max} = h(t_{max}) = K_p\left(1 + e^{-\pi D/\sqrt{1-D^2}}\right)$. The normalized overshoot value $\ddot{u}$ can be determined from the equation $\ddot{u} = \frac{h_{max}-h(\infty)}{h(\infty)} = e^{\frac{-\pi D}{\sqrt{1-D^2}}}$, which can be used to determine the damping ratio D from the overshoot of the step response. Given by the equation $D = \frac{|\ln(\ddot{u})|}{\sqrt{\pi^2+\ln^2(\ddot{u})}}$. new line

Step and Impulse response

explain two more system behaviors

2 Experiment T2 — System identification

2.1. System $N = 0$

2.1.1. Experimentaly Identified system parameters

2.1.2. Documentation of the results

Essential Commands

Differential Equation

Transfer Function

2.1.3. Realworld Example

A practical example of such a PT1 system with impulsive excitation is the temperature response of a small metal body to a brief heat flux. If, for example, a metal rod is briefly excited by a heat source (e.g. hot air or a laser) — comparable to a Dirac impulse — the body heats up abruptly and then cools down again via heat conduction and dissipation to the surroundings. The thermal behaviour follows a PT1 model: the temperature rises briefly due to the pulse (corresponding to the step in the output signal) and then falls back to the original level exponentially with a characteristic time constant. The time constant depends on the material properties (heat capacity, thermal conductivity) and the environmental conditions (e.g. convection, radiation). Such system responses frequently occur in sensor technology, materials analysis or in the case of short control pulses in electronic or thermal systems.description of the respective input and output variables.

2.1.4. Comparison of given System with the response of the identified system

Deviations of the System

2.2. System $N = 1$

2.2.1. Experimentaly Identified system parameters

2.2.2. Documentation of the results

Essential Commands

Differential Equation

Transfer Function

2.2.3. Realworld Example

description of the respective input and output variables.

2.2.4. Comparison of given System with the response of the identified system

Deviations of the System

2.3. System $N = 2$

2.3.1. Experimentaly Identified system parameters

2.3.2. Documentation of the results

Essential Commands

Differential Equation

Transfer Function

2.3.3. Realworld Example

A clear example of an IT1 element with an upstream dead time is a water tank with a constant inflow and proportional outflow. The inflow corresponds to the input signal, and the water level to the output signal. The storage results in integrative behaviour, whilst the proportional outflow causes first-order damping — typical of an IT1 system. In the experimental setup under consideration, a dead-time element was connected upstream of the IT1 element, causing a time delay in the input signal. This means that the system output only reacts to a change after the dead time has elapsed. Such combinations of dead time and IT1 behaviour frequently occur in real-world processes, for example in sluggish piping systems or delayed control variable transmissions. description of the respective input and output variables.

2.3.4. Comparison of given System with the response of the identified system

Deviations of the System

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B.2. Code used for the analysis of the Experiment

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